HUMAIN Future AI Builders Hackathon

HUMAIN Lifecycle

Autonomous Enterprise AI

Turning incoming employment contracts into verified onboarding preparation through autonomous AI.



CHALLENGE 3 Autonomous Enterprise

The enterprise onboarding problem

A single employment contract creates work across disconnected teams.

01 Manual contract review

Read documents and re-enter the same employee facts.

02 Disconnected HR and IT

Coordinate handoffs across separate requests and inboxes.

03 Policy inconsistencies

Check each contract against the right approved rules.

04 Repetitive administration

Create profiles, tasks and welcome messages by hand.

05 Limited workflow visibility

Find out what ran, what failed and who must act next.

06 Deadline monitoring

Track contract, Iqama and work-permit dates.

The coordination burden continues until someone verifies the outcome.

Our solution: verified onboarding preparation

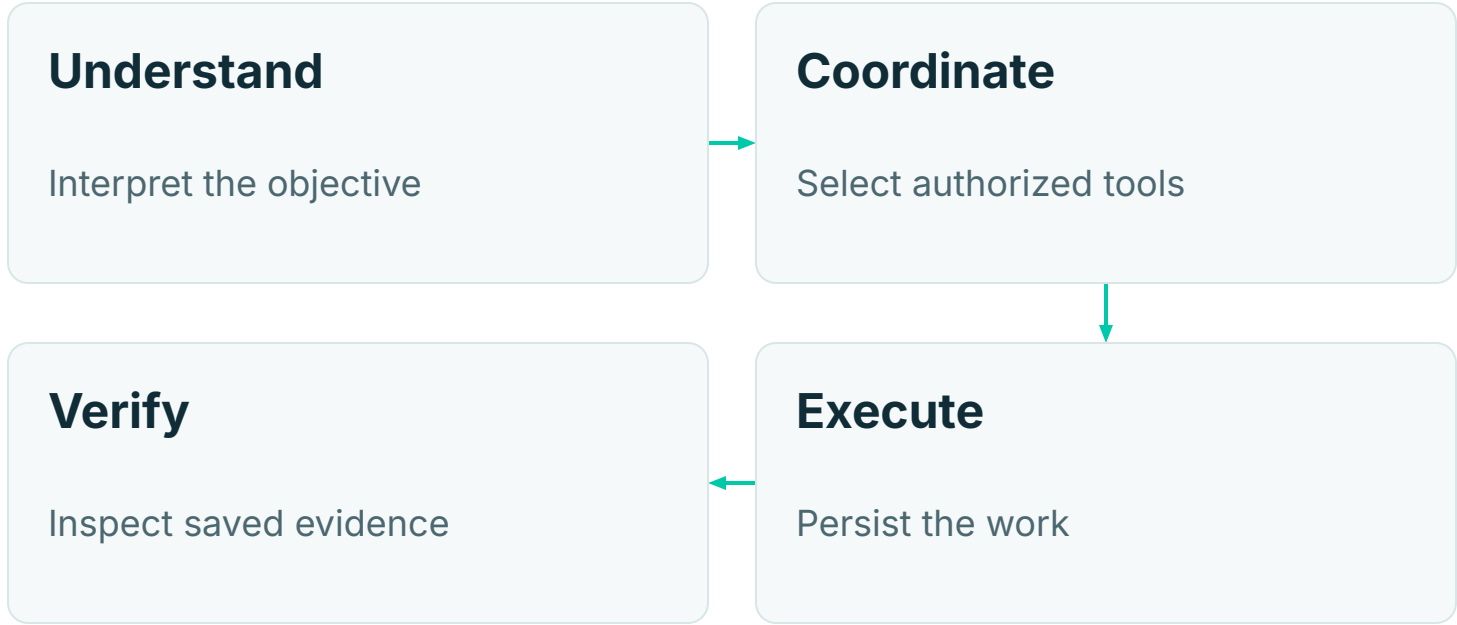
HUMAIN Lifecycle connects an AI-native employee lifecycle workspace to durable HR workflows.

BUSINESS OBJECTIVE

“Prepare onboarding from this contract and verify the result.”

HUMAIN Chat

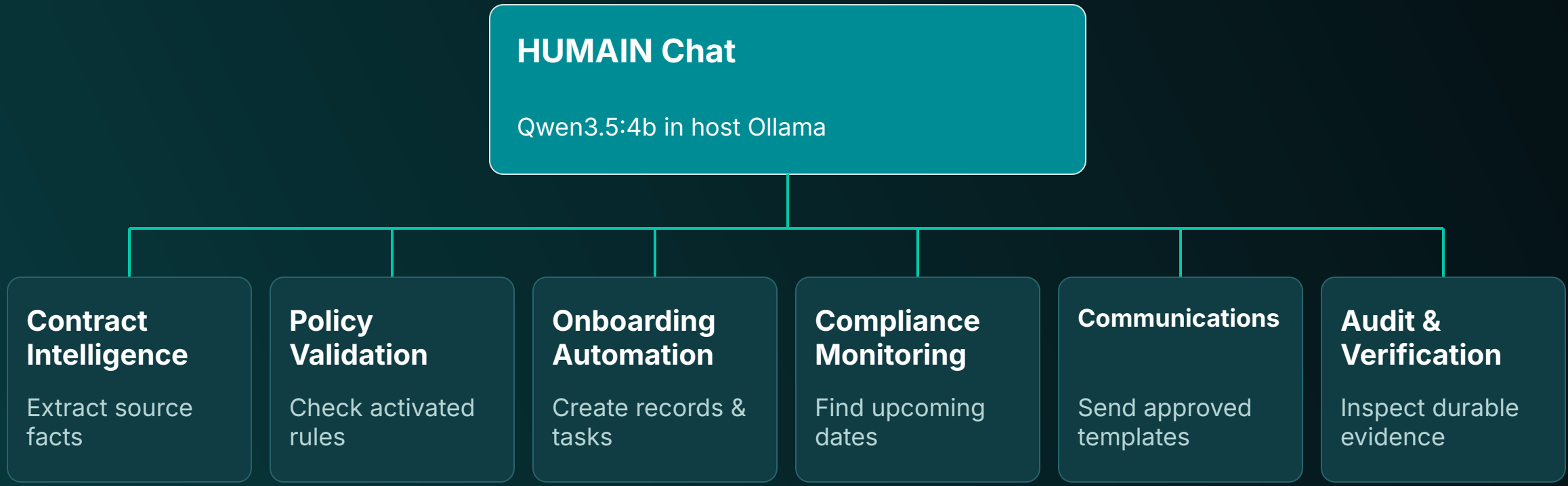
Local Qwen3.5:4b orchestrator



Autonomous preparation	Human approval of identity, employment terms and official Qiwa actions

HUMAIN Chat & specialized capabilities

A local language model coordinates typed tools and backend services.

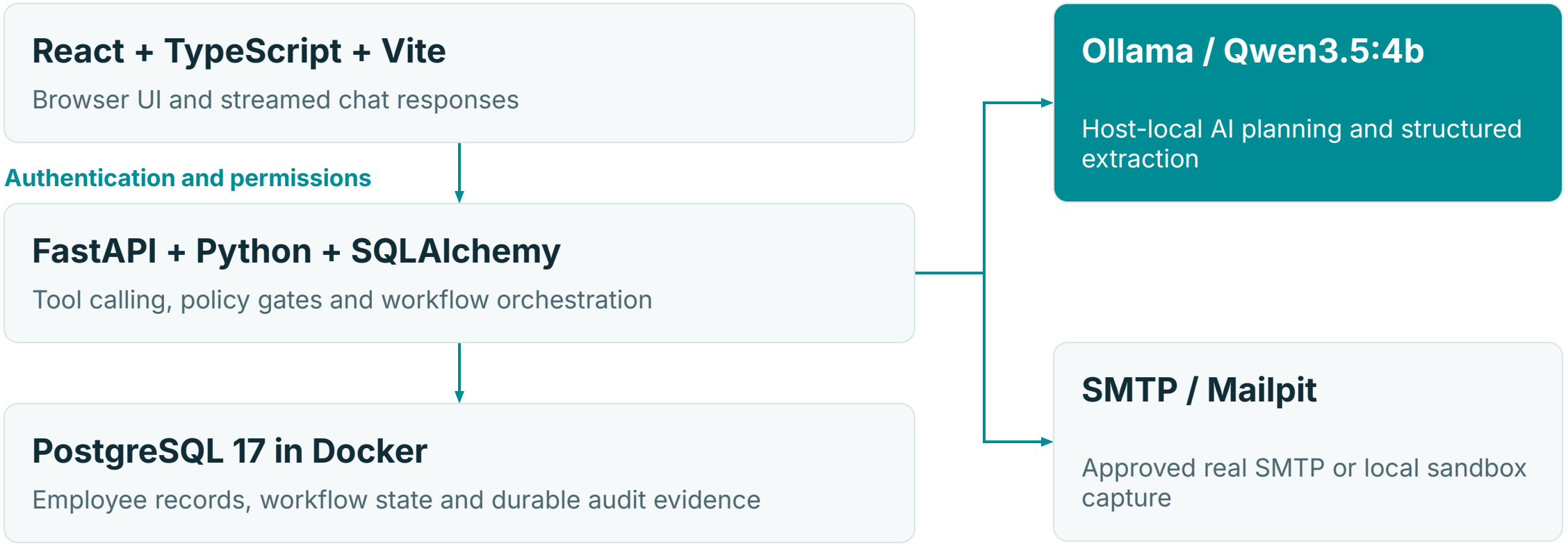


Coordinated AI-enabled tools and backend capabilities

These capabilities do not represent six independent language models.

Technical architecture

Local AI inference with server-enforced permissions and a persistent system of record.

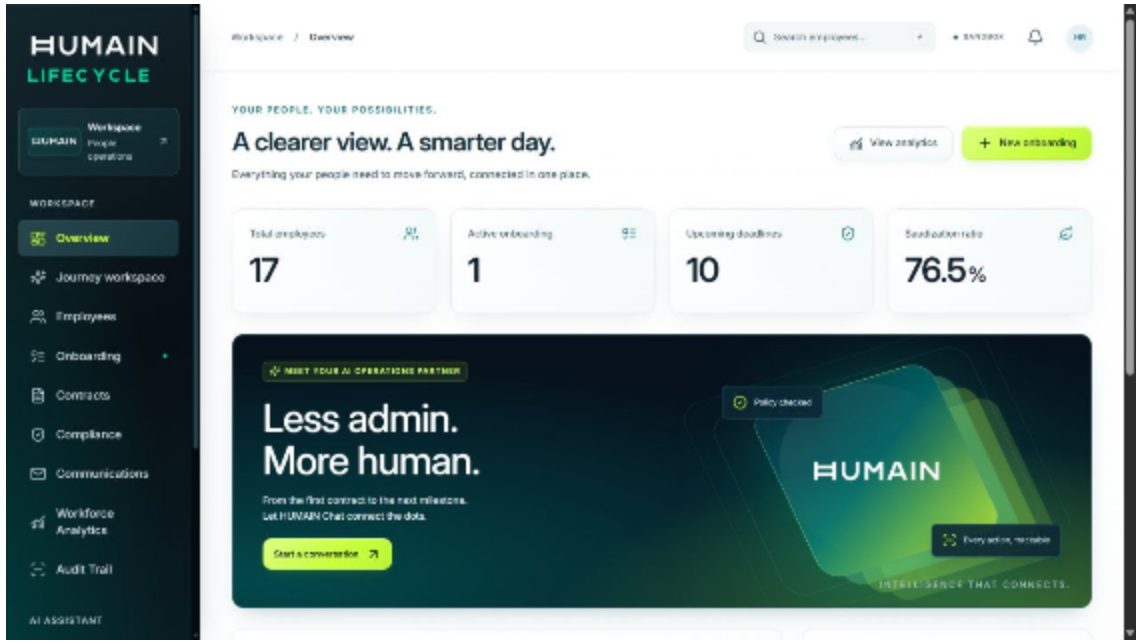


Runtime AI uses local Ollama inference. The implementation does not call the OpenAI API.

Real product implementation

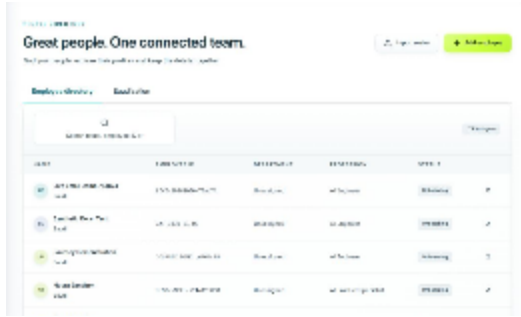
Five views from the running application, using synthetic workspace records.

01 Executive dashboard



Workforce counts, deadlines and recorded activity

02 Employee management



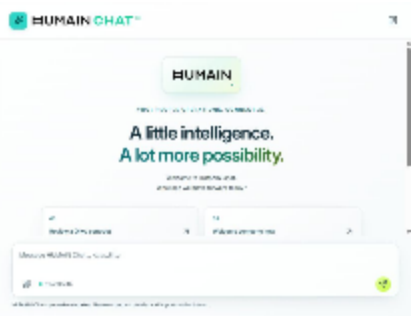
03 Contract intake



04 Policy management

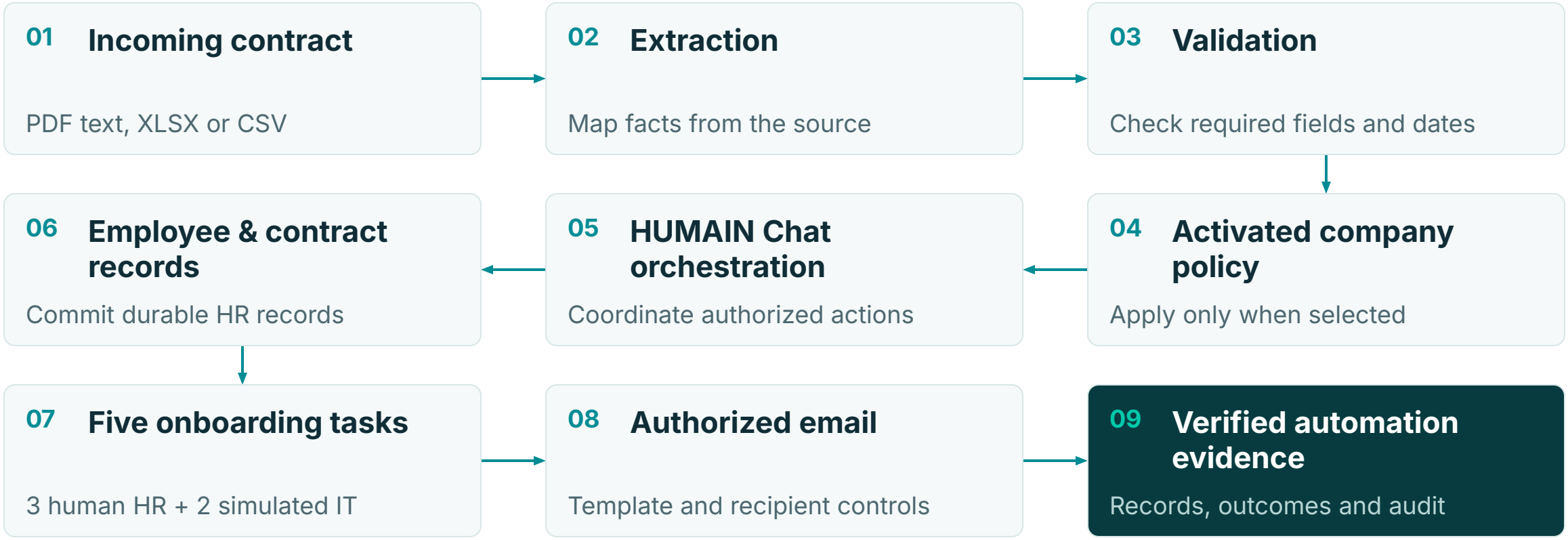


05 HUMAIN Chat



End-to-end autonomous workflow

One contract becomes a persisted objective with evidence and reviewable exceptions.



Exception path: missing facts or policy findings pause execution for authorized review, then resume.

Testing & recorded results

Reproducible checks and one recorded live demonstration.

LATEST RECORDED TEST RUN

101 automated tests passed

81 backend 3 frontend unit 17 browser

Lint, type checking, production build and migration checks passed

Automation setup verified

ONE RECORDED LIVE RUN

37.036 s

Full agent request, including Qwen overhead

2.074 s

Core workflow execution

5 tasks	3 human pending + 2 simulated IT	**SMTP accepted**	0 new attempts on retry

Single demonstration, not an average or success rate. SMTP acceptance does not prove inbox delivery.

Automated test run: 30 Oct 2026. Live-run timings are separate recorded measurements.

Innovation, business value & limitations

A working prototype with explicit boundaries and inspectable outcomes.

WHAT THE PROTOTYPE CONTRIBUTES

Business objectives drive tool orchestration

Locally hosted AI inference

Policy-aware execution with review gates

Durable workflows that resume

Duplicate prevention across retries

Human oversight for consequential decisions

Transparent records and audit evidence

CURRENT LIMITATIONS

Scanned PDF OCR is unavailable

External IT provisioning remains simulated

Human HR approvals remain necessary

Held-out task accuracy and time savings have not yet been established

The prototype is not production-ready

FUTURE WORK

OCR support, real IT connectors, broader held-out evaluation and measured manual baselines

Conclusion & demo

Challenge 3: Autonomous Enterprise

An incoming contract. Verified preparation. Human judgment retained.

HUMAIN Lifecycle prepares onboarding and escalates exceptions that require human judgment.

Verified evidence

Employee record	630421cd-afe3-449e-bed0-25f1135feb8b
Contract record	fadcab75-1404-4a03-b501-e4967b9c60db
Internal tasks created	5
Human tasks remaining	3
IT requests	2 simulated · no accounts provisioned
Welcome email	SMTP accepted · inbox unverified
Automation setup	Verified

Automation setup does not mean the employee is ready. Human checks and real IT provisioning remain separate.

Source fingerprint: 0032ea3f51f651c96ca53237907781cb637491c0359afba63c620a2c0ce96e78

DEMO **Contract intake** / **Journey & tasks** / **Email receipt** / **Audit**